

SyncDrive

Signal-Aware Navigation and Speed Optimization System

Project Report -- Prototype Stage

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October 2026

1. Executive Summary

This report documents the prototype stage of SyncDrive, a signal-aware navigation system that recommends a departure time and a steady driving speed so that a driver traverses an urban corridor while hitting the maximum possible traffic signals on green, eliminating avoidable stop-and-wait time.

The prototype covers seven locations in Chandigarh (Sector 17, 22, 35, 43, PGI Hospital, Elante Mall, Tribune Chowk), models sixteen traffic signals with configurable cycle timings, and provides 42 directed routes. It is a single-file web application requiring no installation, no API key, and no backend server.

Core hypothesis validated: a driver travelling at a lower steady speed with zero red-light stops arrives faster than one rushing at maximum speed with signal waits.

Worked example -- Sector 17 to Sector 22, arrival target 09:00:

Strategy	Speed	Drive time	Wait time	Total
Normal GPS	50 km/h	1.7 min	2.1 min	3.8 min
SyncDrive	32 km/h	2.6 min	0 min	2.6 min

2. Prototype Scope

2.1 What was built

- Interactive dark-theme map -- Leaflet.js with Stadia alidade_smooth_dark tiles (no API key).
- 7 named locations as origin/destination options.
- 16 simulated traffic signals with configurable cycle, green-start offset, and duration.
- 42 bidirectional routes between all location pairs with signals, polyline, and steps.
- Green wave optimizer -- 9 candidate speeds (25-45 km/h), 25-min departure window, 15 s steps.
- Normal GPS simulator -- same departure time, 50 km/h, accumulates real red-light wait time.
- Comparison panel -- total time, time-breakdown bars, red lights hit, time lost, fuel.
- Signal timeline -- per-signal arrival time and green/red status.
- Speed HUD -- map overlay with recommended speed and contextual advice.

2.2 What was deferred

- Real-time GPS position tracking.
- Live municipal signal phase data.
- Animated car movement along route.

- Backend API or database.
- Emergency vehicle coordination.
- Mobile native application.

3. System Architecture

3.1 Repository structure

```
project-root/
  code/
    index.html      <- entire app (map + UI + logic)
    CONTEXT.md      <- handoff documentation
  journals/
    1024030177-tarun/    (5 weekly entries)
    1024030178-parth/    (5 weekly entries)
    1024031077-adhiraj/  (5 weekly entries)
  project-report-prototype-stage/
    main.tex
    main.pdf
```

3.2 Data layer

- PLACES -- 7 locations: name, lat, lng.
- SIGNALS -- 16 objects: id, name, lat/lng, cycle(s), greenStart, greenDuration.
- ROUTES -- 42 directed objects: sigs[], km, pts[][] (polyline), steps[] (directions).

3.3 Green wave optimizer

```
for each (speed in [25,28,30,32,35,38,40,42,45]):
  driveSec = (km / speed) * 3600
  for each offset in 0..1500 step 15:
    dep = arrivalTime - driveSec - offset
    elapsed = 0; totalWait = 0; greens = 0
    for each signal:
      phase = (dep + timeToSignal + elapsed) % cycle
      if red: wait = nextGreen(signal,phase)
          elapsed += wait; totalWait += wait
      else: greens++
    score = (driveSec+totalWait) - 0.5*greens # lower=better
    if score < best: best=score; saveResult()
```

3.4 Normal GPS simulation

Uses the same departure time as SyncDrive but drives at 50 km/h, hitting signals at uncontrolled phases and accumulating real wait times -- ensuring a direct, fair apples-to-apples comparison.

3.5 UI: three-pane layout

Pane	Width	Content
Nav rail	200 px	Logo, Search, Guides, Directions
Directions panel	320 px	Inputs, results, comparison, signal timeline
Map	flex:1	Leaflet map, route polyline, signal markers, speed HUD

#0a84ff (blue) for all interactive elements; #30d158 (green) used only for "signal is green" indicators.

4. Results and Evaluation

4.1 Hypothesis validation across routes

Route	km	NG spd	NG total	GW spd	GW total	Saved
Sec 17 -> Sec 22	1.4	50	3.8m	32	2.6m	1.2m
Sec 22 -> Sec 35	2.2	50	6.0m	35	4.5m	1.5m
Sec 17 -> PGI	3.2	50	5.8m	38	5.0m	0.8m
Sec 43 -> Elante	2.0	50	4.1m	30	4.0m	0.1m
Sec 17 -> Tribune	5.1	50	8.2m	40	7.7m	0.5m

4.2 Key observations

- Shorter routes (1-2 km, 3-4 signals): highest savings ratio as wait/drive ratio is largest.
- Longer routes (5+ km): lower-speed penalty partially offsets savings; SyncDrive still wins or ties.
- Elapsed accumulation is critical -- each red stop shifts all subsequent arrival times.
- Same departure time for both strategies is essential for a fair comparison.

4.3 Limitations

- Signal timings are simulated, not sourced from Chandigarh municipal data.
- Uniform signal spacing assumed per route segment.
- No real vehicle position tracking -- fixed departure + speed, not continuous.
- No user study conducted; figures are computational.

5. Implementation Notes

5.1 Technology decisions

- Leaflet.js 1.9.4 -- no API key, no billing.
- Stadia alidade_smooth_dark tiles -- free tier, dark navy look, no key.
- Single HTML file -- no Node.js, npm, or local server needed.
- Inter (Google Fonts) -- Apple system-font aesthetics, free.

5.2 Bugs fixed during prototype

- W2: Carto tile watermark -- switched to Stadia tiles.
- W3: Wrong optimizer objective -- maximising greens ignores total time; fixed to minimise drive+wait.
- W3: Missing elapsed accumulation -- red stops not shifting downstream signal phases.
- W5: Unfair comparison baseline -- Normal GPS used arrival time as dep reference.
- W5: Missing routes -- 22 of 42 pairs undefined; all now implemented.

6. Next Steps

- Live position tracking via browser Geolocation API.
- "Leave in X minutes" countdown from current system time.
- Animated route playback at the recommended speed.
- Real signal data from OpenStreetMap highway=traffic_signals tags.

- Peak hour signal profiles (morning/evening rush).
- OSRM routing integration for real road-following paths.
- Emergency vehicle coordination (Phase 2 from proposal).

7. Repository

- Code: <https://github.com/ParthPuri/SyncDrive/tree/master/code>
- Journals: <https://github.com/ParthPuri/SyncDrive/tree/master/journals>
- Prototype: [code/index.html](#) -- open in any browser, no server.